

Automatic Relation Extraction from Text: A Survey

Kun Li, Junsheng Zhang*, Changqing Yao, Chongde Shi
 Research Center for Information Science Theory and Methodology
 Institute of Scientific and Technical Information of China
 Beijing, China
 Email: {likun2015, zhangjs, yaocq, shicd}@istic.ac.cn

Abstract

Relation extraction is an important task for understanding text. In the big data era, automatic relation extraction from unstructured texts is urgently needed for structured information organization and information analysis. In this paper, we survey the automatic relation extraction methods, especially the traditional machine learning on closed data set and open information environment such as Web, including supervised and semi-supervised methods. And then, we discuss the applications based on relation extraction such as event extraction and QA systems.

Index Terms

Relation extraction; Machine learning; Information extraction

I. INTRODUCTION

During the search on the Web, relations between entities are interesting which are shown as relation-related queries such as *people-affiliation* and *organization-location*. In the existing research work, NER (Named Entity Recognition) has been used to automatically extract entities with high accuracy, but the relation between entities is not easy to identify and extract. In the early stage, tuple $(e_1, e_2, e_3, \dots, e_n, r)$ is used to represent a relation group, in which $e_i (i = 1, \dots, n)$ represents an entity in the relational group and r means the relations between entities [1]. Therefore, the research interests have been changed to extract binary relations which is unrepresented by a tuple (e_1, e_2, r) in which r could be specific relations such as *author-book* relation (DIPRE), *organization-location* relation (snow ball), and *person-affiliation* relation (tree kernel).

MUC(Message Understanding Conferences) and ACE (Automatic Content Extraction) meeting are two most important meetings in information extraction research field, which have greatly promoted the development of relation extraction. The development of the relation extraction methods has three stages: rule-based information extraction, traditional machine learning relation extraction and open information extraction (Open IE).

- 1) Rule-based relation extraction. Before statistical machine learning method is used to extract relation, relation extraction rules are developed by domain experts, and then relations are extracted by computer according to these rules. The rule based extraction system such as [2] is highly dependent on the rules, so the performance of rule generalization is poor, and the rules are either too simple or too complex in applications. Therefore, the rule based relation extraction method is not very popular because of the adaptiveness.
- 2) Traditional machine learning based relation extraction. With the rapid development of machine learning, many Natural Language Processing (NLP) research fields including relation extraction began to use machine learning algorithm. In the earlier stage, there are only a small amount of machine learning based systems such as Ref. [3]. However, machine learning based system are shown to be better than the traditional rule based system in precisiongeneralization performance and efficiency. So machine learning method became the dominate method of relation extraction and replaced the rule-based relation extraction method.
- 3) Open IE. With the development of the Internet and Web, text information on the Internet has become the main target of relation extraction. Internet contains massive and complex data. How to make use of these unlabeled unstructured text and improve the generalization of the relation extraction system on the large scale document collection becomes a challenge for the relation extraction task. And then, many unsupervised and semi-supervised machine learning methods have been proposed to fulfill the open information extract task. In addition, the distance supervised method based on knowledge bases like Freebase and Wiki is also worthy of attention.

In this paper, we focus on traditional supervised machine learning methods and Open IE by using semi-supervised methods.

II. MACHINE LEARNING BASED RELATION EXTRACTION

The basic idea of the traditional machine learning method is to transform the relation extraction problem into a binary classification problem, and then train the supervised system with labeled text [4]. Further, the supervised methods of relation extraction include features based method and kernel method. Feature based method is needed to manually set the feature-matrix of text, while kernel method does not manually labeling. Feature based method may have better results in specific areas.

The process of dealing with the problem of relation extraction with machine learning method contains the following steps:

- 1) Find sentences that contain relations and entities in the texts. Giving a sentence $S = group_1(ws_1 e_1 ws_2 e_2 ws_3, r_1), group_2(ws_1 e_1 ws_2 e_2 ws_3, r_2), \dots$, where e_1 and e_2 are entities, ws_1, ws_2 and ws_3 are the word sequences around the entities, and r_1 and r_2 are the relations between e_1 and e_2 .
- 2) Transform relation groups into corresponding feature matrix, that is, $F(group)$.
- 3) Set a hypothesis $h(F(group))$, if the value of $h()$ is positive then it means that the group implies relation r , otherwise does not imply relation r .

In this way, the task of extracting relation are turned into the task of identifying a specific relation in a sentence. With a number of text with *positive* and *negative* labels as the training data, the supervised relation extraction system can be constructed by using SVM, Logistic or other binary classifiers.

A. Feature-based Method

In feature-based method, it is necessary to manually set the feature matrix. Features can be classified into two types: *semantic feature* and *syntactic feature*. The syntactic features are extracted from sentence directly, including the entities, the types of the entities, the word sequence between the entities, the number of words between the entities and the path in the parse tree containing the two entities.

The core issue of extracting relation based features is to choose the proper sentence features. It is difficult to find the most suitable features for extracting specific relation from specific training samples. Therefore, the selection of sentence feature can affect the effect of relation extraction. The text with different structures and different types of entity relation have completely different sensitivity for features, so in dealing with a specific relation extraction problem must be chosen repeatedly until the most appropriate feature space is found.

B. Kernel Methods

As the above mentioned, feature-based classification has some limitations. There are a variety of supervised methods based on kernels to select proper features. String Kernel is the basis of all kinds of kernel method, so it is very important to deeply understand the String Kernel method.

The essence of String Kernel [5] is to calculate the similarity between two strings, whose basic idea is as follows: Given two strings x and y , the more x and y have common substrings, the higher the similarity is. The specific method is to map the compared strings into a high dimension feature space. The dimensions of feature space are potential substrings of the string.

The value of a dimension may be 0 (without that substring), or be the weight of substring. For example, if string x is “dog” and y is “dig”, they are mapped to a high dimensional space.

Intuitively, if a substring appears in the string, the corresponding dimension of value is the power function value of the decay factor, and the power exponent decided by the span in the string. Suppose the length of $k = 1$, the power exponent is 1. For the substring whose length is 2, if the substring is continuous in the string, power exponent value is 2; If it is non-continuous, the exponent is greater than 2.

Suppose Σ is a finite alphabet set and it contains all the letters in the string. A string is a finite sequence composed by the letters in Σ , which can be an empty string. Suppose there are two strings s, t that satisfy the above conditions. Using $|s|$ to present the length of the string s , so $s = s_1, \dots, s_{|s|}$. If there exist indices $I = (i_1, \dots, i_u)$, u is used to represent a substring in s , and $u = s[I]$ for short. Substrings $u_j = s_{i_j}$ and $L(i) = i_{|u|} - i_1 + 1$ represent the span in the string. Σ^n is used to represent all the strings whose lengths equal to n and composed by the letters in set Σ . For all $u \in \Sigma^n$ which means the substrings with n length, we can map the strings into u feature space. Then the String Kernel value can be calculated for strings with length n . Finally, the String Kernel formula is shown as follows:

$$K_n(s, t) = \sum_{u \in \Sigma^n} \sum_{i: u=s[i]} \sum_{j: u=t[j]} \lambda^{l(i)+l(j)}$$

With the method, we can find whether the target string is similar to the training samples having a labeled relation, and then corresponding relation can be found.

III. OPEN INFORMATION BASED RELATION EXTRACTION

Relation extraction systems use traditional supervised learning method to extract relations. The main problem is that these systems are difficult to be obtained with a large number of labeled training data. ACE conference provides 17,000 training data for relation extraction problem. However, such a small number of training samples in the extraction of a variety of complex type of relation is very difficult to get a good result [6]. Facing the requirement of Internet information extraction application, researchers begin to transfer the research object from the traditional limited texts to the open information on the Internet. The goal is to use the massive, redundant, unstructured texts on the Internet which contains a lot of noise to train the relation extraction model, so we can use the system to extract new unstructured texts. For relation extraction from open information, the text types are variety; relation types are variety; and, the characteristics are variety.

Supervised machine learning based relation extraction method uses a lot of labeled data for training, which has achieved good results in some cases, especially compared with the rule-based methods. However, there are some problems including the lack of training data and low generalization performance. To solve these problems, researchers try to use the vast amount of unlabeled text in the Internet as the training data, and proposed semi-supervised learning algorithms respectively. The basic idea of the algorithms is bootstrapping which uses the output of learning as the training data for the next iteration. DIPRE is a representative relation extraction algorithm, and Open IE research group developed some useful extraction systems based on the bootstrapping method.

A. DIPRE

DIPRE (Dual Iterative Pattern Relation Expansion) is a semi-supervised relation extraction algorithm proposed in 1998. Take *Author-Book* relation as example, the relation extraction steps are as follows[7].

First, the algorithm begins with a small set of *Author-Book* relation pairs as the seed. For simplicity, it is assumed that there is only one relation seed “J.K.Rowling - Harry Potter and the Sorcerer’s Stone”. Next, the algorithms try to find all occurrences of tuples of *Author-Book* relation in the dataset as follows:

“you can read Harry Potter and the Sorcerer’s Stone written by J.K.Rowling online”

“know that J.K.Rowling wrote Harry Potter and the Sorcerer’s Stone in London”

Then, frequent patterns are selected according to the occurrences in the dataset. The definition of a pattern has great influence on the result of algorithm. We can define a simple pattern for test, for example, [order, prefix, author, middle, book, suffix]. There are six features of pattern including the occurrence order of the two entities, author entity, book entity, some strings before entity, some strings after entity and strings between two entities. And then, the above occurrences of tuples are translated into the patterns as follows:

[0, J.K.Rowling, Harry Potter and the Sorcerer’s Stone, Read, online, written by]

[1, J.K.Rowling, Harry Potter and the Sorcerer’s Stone, know that, in London, wrote]

The patterns are generated into groups, which means that each seed works on the limited texts of dataset to extract some occurrences of tuples. For each group of patterns, we could choose the longest common prefix of suffix strings and the longest common suffix of prefix strings to replace the feature of prefix and suffix.

[order, longest-common-suffix of prefix, author, middle, book, longest-common-prefix of suffix strings].

The new patterns generated by tuple groups are used to search tuples in dataset matching the patterns.

B. Open Information Extraction

Open IE is developed by the University of Washington. The system runs over sentences and creates extractions that represent relations in text. There have been four versions of Open IE system based on bootstrapped method.

- 1) Open IE 1.0 is known as TextRunner, which makes a single data-driven pass over its corpus and extracts a large set of relational tuples without requiring any human input. It is a fully implemented, highly scalable Open IE system where the tuples are assigned a probability and indexed to support efficient extraction and exploration via user queries [8].
- 2) Open IE 2.0 is known as ReVerb. The ReVerb extractor is a novel open system based on some constraints on relation phrases. It differs in three important ways from previous methods like TextRunner. The relation phrase is identified “holistically” rather than word-by-word. Potential phrases are filtered based on statistics over a large corpus. REVERB is “relation first” to avoid to confuse a noun in the relation phrase for an argument [9].
- 3) Open IE 3.0 used to be known as Ollie. ReVerb system has two important weaknesses. First, it extracts only relations that are mediated by verbs; Second, it ignores context, and thus it extracts tuples that are not asserted as factual. Ollie system is designed to address these two limitations. First, Ollie achieves high yield by extracting relations mediated by nouns, adjectives, and more. Second, a context-analysis step increases precision by including contextual information from the sentence during the extractions [10].
- 4) Open IE 4.0 is composed of Srlie [11] and Relnoun [12], which improved previous systems by using hybrid method. There are a lot of relation data extracted by Open IE and some applications based on the system. For example, the Paralex [13] is a system that learns to answer questions and the Semantic Scholar (www.semanticscholar.org) which is an AI-based academic search engine.

IV. DISCUSSION

Relations extracted from texts could be used in event extraction and QA systems.

- *Event extraction.* In big data era, massive data is generated online every day. Users need to extract complex event from the cross-media information on Internet [14]. Given unstructured dataset such as web documents and texts, it is necessary to extract entities, relations and other arguments that are implicit semantic attributes of data and organize the extracted information into structured models such as Event Cubes [15] and temporally linked network [16]. Besides, event can be recommended after extracting events and computing similarity between the historical events and newly happened events.

Event extraction techniques can be applied to various applications including event tracking, providing alerts and predicting changes.

- *QA system.* It is difficult for current information retrieval systems to return intelligent retrieval result for queries represented in nature language. So QA systems will become the main form of the future retrieval systems [18]. For a query like “What year was Gandhi born” and “How long is Yao Ming”, it needs to recognize the entities and relations in texts. With a new round of research in Artificial Intelligence, QA system research becomes hot topic in research fields such as knowledge engineering and natural language processing.

Although the machine learning method is the main method in relation extraction research field, there are still many aspects to be improved [19]. Besides, machine learning method of relation extraction also needs to be integrated with other methods.

V. CONCLUSION

In this paper, we survey the popular methods of relation extraction including rule-based relation extraction, machine learning based relation extraction and open information based relation extraction. Machine learning based information extraction has replaced rule-based method and become the dominant method for relation extraction. However, machine learning methods still have many shortcomings and limitations, such as the most of relation extraction systems focus on binary relation extraction. It is a trend to extract n -ary relations on a large scale data. Besides, relation extraction systems that do not require large-scale training data will be more popular in the future with some new algorithms such as Text Runner [20]. Relation extraction has been an important basic technology for constructing QA systems and new intelligent applications such as information analysis.

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